

# Summary and Recommendation

## 1. Business Overview

This project analyzes customer churn in a telecommunications company to understand the characteristics and service-related factors associated with customers leaving the company. The analysis uses **7,043 customer records** and explores churn from multiple dimensions, including customer demographics, tenure, contract type, internet and additional services, and payment methods.

The analysis found that **26.54% of the customers have churned**, meaning approximately **one out of every four customers** has discontinued the company's services. This represents a significant customer-retention challenge and highlights the need to identify high-risk customer segments and develop targeted retention strategies.

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## 2. Overall Churn Situation

The overall churn distribution provides the most important high-level business insight:

- **26.54% of customers have churned.**
- Approximately **73.46% of customers have remained with the company.**
- Therefore, the company is retaining nearly three-quarters of its customer base, but losing more than one-quarter of customers represents a substantial potential loss of recurring revenue.

The relatively high proportion of churn indicates that customer retention should be treated as an important business priority.

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## 3. Customer Demographics and Churn

### Gender

The analysis compares churn across male and female customers. The visualization does not indicate a major difference in churn based on gender.

This suggests that **gender is not a particularly strong factor for identifying customers at risk of churn**. Therefore, retention strategies should not rely heavily on gender-based segmentation.

## Senior Citizens

The analysis shows a **comparatively greater percentage of senior citizens have churned** compared with non-senior customers.

This indicates that senior citizens may represent a relatively higher-risk customer segment. The company could therefore consider providing more personalized customer support, simplified service plans, and targeted engagement programs for this group.

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## 4. Tenure – One of the Most Important Findings

Customer tenure shows a clear relationship with churn.

The analysis indicates that:

- Customers with **very short tenure, particularly around the first one or two months, show substantially more churn**.
- Customers who have stayed with the company for a longer period are considerably more likely to remain.
- Churn appears to decrease as customers become more established with the company.

This is an important business insight because it suggests that the **early customer lifecycle is a critical retention period**.

The company should therefore focus on the first few months after acquisition by providing:

- Better onboarding experiences
- Early customer satisfaction checks
- Promotional offers for new customers
- Proactive technical support
- Personalized communication
- Incentives to move customers toward longer-term contracts

Improving retention during the early months could have a significant impact on the overall churn rate of **26.54%**.

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## 5. Contract Type and Churn

Contract type is one of the clearest patterns identified in the analysis.

Customers with **month-to-month contracts are considerably more likely to churn** than customers with one-year or two-year contracts.

In comparison, customers who commit to longer-term contracts demonstrate substantially stronger retention.

This suggests that contract commitment is strongly associated with customer loyalty. From a business perspective, the company could encourage month-to-month customers to move toward one-year or two-year contracts through:

- Contract upgrade discounts
- Loyalty benefits
- Reduced monthly pricing
- Additional services bundled with longer contracts
- Special offers for customers who renew

The month-to-month customer segment should therefore be considered a **high-priority churn-risk group**.

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## 6. Internet and Additional Services

The service-level analysis provides several important insights.

### Internet Service

**Fiber Optic customers show noticeably higher churn**, while customers using **DSL show comparatively lower churn**.

This is an important finding because fiber optic customers may have higher expectations regarding speed, reliability, service quality, or pricing. The company should investigate whether the higher churn is related to:

- Pricing
- Service quality
- Technical problems
- Customer expectations
- Competitor offerings
- Perceived value for money

## Additional Services

Customers who do **not** subscribe to services such as:

- Online Security
- Online Backup
- Device Protection
- Tech Support

show relatively higher churn compared with customers who have these services.

This suggests that additional services may be associated with stronger customer retention.

The company could therefore consider bundling these services with internet plans or offering personalized discounts to customers who currently do not use them.

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## 7. Payment Method

Payment method is another important pattern identified in the analysis.

Customers using **Electronic Check** appear more likely to churn compared with customers using other payment methods.

This makes electronic-check customers an important segment for further investigation.

The company could examine whether this relationship is associated with:

- Payment convenience
- Failed or delayed payments
- Customer preferences
- Billing experience
- Lack of automatic payment
- Customer demographics associated with this payment method

Encouraging customers toward more convenient and automated payment options may potentially improve retention, although further analysis would be required to establish whether payment method directly affects churn.

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## 8. Key Churn Risk Segments

Based on the exploratory analysis, the following customer groups appear to require greater attention:

| Customer characteristic  | Churn implication              |
|--------------------------|--------------------------------|
| Overall customer base    | <b>26.54% churn rate</b>       |
| New/low-tenure customers | Higher churn risk              |
| Month-to-month contracts | Higher churn risk              |
| Senior citizens          | Comparatively higher churn     |
| Fiber Optic customers    | Noticeably higher churn        |
| No Online Security       | Relatively higher churn        |
| No Online Backup         | Relatively higher churn        |
| No Device Protection     | Relatively higher churn        |
| No Tech Support          | Relatively higher churn        |
| Electronic Check users   | Higher churn tendency          |
| Gender                   | No major difference identified |

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## 9. Overall Business Interpretation

The analysis suggests that customer churn is not evenly distributed across the customer base. Instead, certain customer characteristics are associated with substantially greater churn risk.

The strongest patterns observed are related to **customer tenure and contract type**, with new customers and month-to-month customers appearing particularly vulnerable. Service characteristics also matter: **Fiber Optic customers and customers without several value-added services show higher churn tendencies**. In addition, **senior citizens and Electronic Check users** represent segments that may deserve additional attention.

With an overall churn rate of **26.54%**, the company has a significant opportunity to improve customer retention by identifying high-risk customers before they leave.

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## 10. Recommended Business Actions

Based on the findings, the company should consider the following strategies:

### **1. Strengthen New-Customer Retention**

Focus heavily on customers during their first few months, since the analysis shows substantially higher churn among low-tenure customers.

### **2. Convert Month-to-Month Customers**

Provide incentives for customers to shift from month-to-month contracts to one-year or two-year contracts.

### **3. Investigate Fiber Optic Churn**

Conduct a deeper investigation into pricing, service quality, technical complaints, and customer satisfaction among Fiber Optic users.

### **4. Promote Value-Added Services**

Offer bundled or discounted packages containing Online Security, Online Backup, Device Protection, and Tech Support.

### **5. Target High-Risk Payment Segments**

Investigate why Electronic Check users show higher churn and encourage convenient automated payment options where appropriate.

### **6. Develop Senior-Citizen Retention Programs**

Provide personalized support and simplified plans for senior customers who demonstrate comparatively higher churn.

### **7. Build a Churn Prediction System**

The exploratory analysis identifies important patterns, but the next step should be to build a **machine-learning churn prediction model**. This would allow the company to assign a churn probability to each customer and proactively contact customers with the highest predicted risk.

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## **Conclusion**

The Telecom Customer Churn analysis reveals that **26.54% of the customer base has already churned**, making retention a major business concern. The analysis shows that churn is

particularly associated with **short customer tenure, month-to-month contracts, senior-citizen customers, Fiber Optic internet service, lack of additional services, and Electronic Check payment methods.**

The findings suggest that the company should move from a general retention strategy toward a **targeted, data-driven retention approach.** Prioritizing new customers, month-to-month subscribers, and other high-risk segments could help reduce churn and protect recurring revenue.

Overall, the analysis demonstrates how exploratory data analysis and visualization can transform raw customer data into actionable business insights. The next logical step is to use these identified patterns to develop a **customer churn prediction model**, enabling the company to intervene before high-risk customers actually leave.